

Self-Study Dashboard

Use-Case Specification: UC6 - Multiple Choice

Version: 1.0

Revision History

Date	Version	Description	Author
23/Jul/26	1.0	Initial draft for UC6: Multiple Choice	System Analyst

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1. Use-Case Name

UC6: Multiple Choice

1.1 Brief Description

This use case describes the specific interaction when a Learner encounters a Multiple Choice question during a quiz. As a specialization (generalization relationship) of UC5: Take Quiz, it details how the system presents a question with predefined options and how the Learner selects their intended

answer(s). It supports both single-correct-answer (radio buttons) and multiple-correct-answer (checkboxes) formats.

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Question 4 of 1040%

B1 Intermediate

Which sentence correctly uses the Present Perfect for a life experience?

☐ A) 'I have visited Paris in 2010.'

☐ B) 'I have visited Paris twice.'

☐ C) 'I visit Paris twice.'

☐ D) 'I am visiting Paris twice.'

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Check Answer ✓

Figure 6.1: Multiple Choice question (single-answer, radio button format) showing four answer options with a "Check Answer" action.

2. Flow of Events

2.1 Basic Flow

1. The use case begins when the quiz engine (UC5) loads a question designated as the "Multiple Choice" type.
2. The system displays the question prompt text and/or media (images, audio).
3. The system displays a list of predefined answer options below the prompt.
 - If the question allows only **one** correct answer, the options are presented as mutually exclusive radio buttons.
 - If the question allows **multiple** correct answers, the options are presented as independent checkboxes.
4. The Learner selects their desired option(s) by clicking or tapping on them.
5. The system visually highlights the selected option(s) to confirm the input.
6. The system temporarily stores the Learner's selection in the active quiz session.

7. The use case ends when the Learner navigates to another question or proceeds to submit the quiz (returning control to UC5).

2.2 Alternative Flows

2.2.1 Changing Selection

If the Learner changes their mind before navigating away or submitting:

1. **Single-answer format:** The Learner selects a different radio button. The system automatically deselects the previously chosen option and highlights the new one.
2. **Multiple-answer format:** The Learner clicks an already selected checkbox to deselect it, or clicks unselected checkboxes to add to their choices. The system updates the visual state and session storage accordingly.

2.2.2 Maximum Selection Limit Reached (Multiple Answers)

If a multiple-answer question has a predefined maximum number of allowed selections (e.g., "Select the 2 correct statements"):

1. The Learner selects the maximum allowed number of options.
2. If the Learner attempts to select an additional option, the system ignores the input and briefly displays a tooltip or message stating, "You can only select up to [X] options."
3. The Learner must deselect a currently chosen option before selecting a new one.

3. Special Requirements

3.1 Accessibility Requirement

The multiple-choice interface must be fully navigable using a keyboard (e.g., using **Tab** to move between options and **Space** to select/deselect). The options must also be readable by standard screen readers.

3.2 Visual Distinction

The user interface must clearly differentiate between single-answer questions (using standard circular radio buttons) and multiple-answer questions (using standard square checkboxes) so the Learner immediately understands the expected interaction.

4. Preconditions

4.1 Question Instantiation

The Learner must be actively executing UC5 (Take Quiz), and the system must have successfully retrieved a question payload from the database explicitly flagged as a "Multiple Choice" type.

5. Postconditions

5.1 Answer Recorded

The state of the system is updated to reflect the Learner's selected options for this specific question within the temporary quiz session memory. The system is ready to evaluate these answers once the overall quiz is submitted in UC5.

6. Extension Points

6.1 None

There are no extension points for this specific use case.